

PROFESSIONAL INSTALLATION MANUAL

Ubuntu Server Deployment and Initial Configuration

ITEP 414 - System Administration and Maintenance

Week 3 Portfolio Project

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1. Introduction

This professional installation manual provides a step-by-step procedure for deploying Ubuntu Server in an Oracle VirtualBox virtual machine. It is designed so that another system administrator can reproduce the installation without needing additional instructions.

The manual covers virtual machine preparation, Ubuntu Server installation, network and storage configuration, user account creation, hostname configuration, OpenSSH Server installation, initial server configuration, verification procedures, troubleshooting, and basic server administration best practices.

For this deployment, the server is configured with the hostname **server01** and uses DHCP networking through the VirtualBox NAT network. OpenSSH Server is also installed to support secure remote administration.

The completed server can later be prepared for services such as file sharing, remote administration, database hosting, web hosting, and other internal services.

2. System and Software Requirements

2.1 Required Software

Software	Purpose
Oracle VirtualBox	Creates and manages the virtual machine.
Ubuntu Server LTS ISO	Installation media for Ubuntu Server.

A stable internet connection is recommended because Ubuntu may download package information and security updates during and after installation.

2.2 Virtual Machine Specifications

Component	Required Configuration
VM Name	Ubuntu-Server-Week03
RAM	4 GB / 4096 MB
CPU	2 Virtual Processors
Storage	40 GB Virtual Disk
Network	NAT
Optical Drive	Ubuntu Server LTS ISO
Hostname	server01
Network Configuration	DHCP
Remote Administration	OpenSSH Server

3. Ubuntu Server Installation Guide

3.1 Create the Virtual Machine

1. Open **Oracle VirtualBox**.
2. Click **New**.
3. Enter the VM name **Ubuntu-Server-Week03**.
4. Select the downloaded Ubuntu Server LTS ISO.
5. Use a manual installation rather than an unattended installation.

6. Configure 4096 MB RAM, 2 virtual processors, and a 40 GB virtual hard disk.
7. Complete VM creation.

Before starting, verify that the Ubuntu Server ISO is attached to the virtual optical drive.

3.2 Configure the Network

Open **Settings - Network - Adapter 1**. Make sure **Enable Network Adapter** is enabled and **Attached to** is set to **NAT**. NAT allows the VM to access external networks through the host computer.

3.3 Start the Ubuntu Server Installer

Select **Ubuntu-Server-Week03** and click **Start**. When the installer loads, select **Ubuntu Server**. Do not select the minimized installation.

3.4 Select the Language

Set the installation language to **English** and select **Done**.

3.5 Configure the Keyboard

Use **English (US)** for the keyboard layout and variant, then select **Done**.

3.6 Configure the Network

Use **DHCP**. Do not manually configure a static IP address unless required by the network administrator. A VirtualBox NAT environment may assign an address similar to **10.0.2.15/24**. Record the assigned address.

3.7 Configure the Proxy

If no organizational proxy is required, leave the proxy address blank and select **Done**.

3.8 Configure the Ubuntu Archive Mirror

Keep the default Ubuntu archive mirror unless the organization provides a specific mirror. Allow the installer to test it, then select **Done**.

3.9 Configure Storage

Select **Use an entire disk** and choose the 40 GB virtual disk. LVM may remain enabled. Leave LUKS encryption disabled and do not select a custom storage layout. Review the proposed partitions and confirm the operation. The deployment should show an **ext4** filesystem.

Warning: Formatting removes existing data from the selected disk, so verify the correct virtual disk before continuing.

3.10 Configure the User Account and Hostname

Create a non-root administrative account. Set the server name to **server01**. Use the administrator's full name, a suitable username, and a strong password. Do not expose the password in screenshots or documentation.

3.11 Ubuntu Pro

If Ubuntu Pro is offered and is not required by the organization or project, continue without enabling it.

3.12 Install OpenSSH Server

Enable **Install OpenSSH Server**. Do not import an SSH identity unless an approved key is available and required. OpenSSH supports secure remote command-line administration.

3.13 Additional Packages

Do not select additional packages unless they are specifically required.

3.14 Complete the Installation

Allow Ubuntu Server to finish installing the operating system, OpenSSH Server, system packages, and available security updates. Do not forcefully shut down the VM. When installation completes, select **Reboot Now**. Remove the Ubuntu Server ISO if requested so the VM boots from the installed virtual disk.

4. Initial Server Configuration

After rebooting, Ubuntu Server should display a login prompt similar to **server01 login:.** Enter the administrative username and password. Linux normally displays no characters, dots, or asterisks while a terminal password is being typed.

4.1 Verify the Hostname

```
hostname
```

Expected output: **server01**.

4.2 Check the IP Address

```
ip addr
```

Locate the active interface, such as **enp0s3**, and record the IPv4 address.

4.3 Test Internet Connectivity

```
ping -c 4 google.com
```

Successful responses indicate working network connectivity and name resolution for the tested hostname.

4.4 Update Package Information

```
sudo apt update
```

Refreshes information about packages available from configured Ubuntu repositories.

4.5 Upgrade Installed Packages

```
sudo apt upgrade -y
```

Installs available upgrades; **-y** automatically confirms the operation.

4.6 Verify OpenSSH Server

```
systemctl status ssh
```

Look for **Active: active (running)**. Press **q** to exit the status screen.

5. Server Verification Procedures

Test	Command	Successful Result
User Login	Login using administrative account	Login succeeds
Hostname	hostname	server01
IP Address	ip addr	IPv4 address displayed
Internet Connectivity	ping -c 4 google.com	Four responses received
Repository Update	sudo apt update	Repository information retrieved
Package Upgrade	sudo apt upgrade -y	Upgrade completes successfully

Test	Command	Successful Result
SSH Service	systemctl status ssh	active (running)

5.1 Deployment Verification Checklist

- Ubuntu Server successfully installed
- Hostname configured as server01
- Administrative user created
- Strong password configured
- 40 GB virtual disk configured
- Network interface functioning
- DHCP address assigned
- Internet connectivity verified
- Ubuntu package information updated
- Available package upgrades processed
- OpenSSH Server installed
- SSH service active and running
- Installation ISO removed after installation
- Server successfully boots from the virtual hard disk

6. Troubleshooting Guide

6.1 Unable to Boot Ubuntu Installer

Symptoms: The VM displays **No bootable medium found**.

Possible cause: The Ubuntu Server ISO is not attached.

Solution: Power off the VM, open VirtualBox **Settings - Storage**, attach the Ubuntu Server ISO to the optical drive, save, and start the VM again.

6.2 VM Returns to the Installer After Installation

Possible cause: The Ubuntu Server ISO is still being used as boot media.

Solution: Remove the ISO from the virtual optical drive and restart the VM. Do not remove the virtual hard disk.

6.3 Unable to Log In

If Ubuntu displays **Login incorrect**, verify the username, password, Caps Lock state, and typing. Remember that terminal passwords are not visually displayed.

6.4 No IP Address

Verify that VirtualBox Adapter 1 is enabled and attached to NAT. Restart Ubuntu if necessary and run **ip addr** again.

6.5 Unable to Access the Internet

Run **ip addr** first. Verify an IPv4 address is assigned and NAT is enabled. Retry **ping -c 4 google.com**. If an IP exists but hostname-based ping fails, DNS may require investigation.

6.6 APT Update Fails

Possible causes include no internet connectivity, repository connectivity problems, DNS issues, or incorrect network configuration. Test connectivity before retrying **sudo apt update**.

6.7 SSH Service Is Not Running

Check **systemctl status ssh**. If installed but stopped, run **sudo systemctl start ssh** and verify again.

6.8 Virtual Machine Becomes Slow

Close unnecessary host applications and avoid running multiple resource-intensive VMs simultaneously. Do not force-shutdown the VM merely because an installation or update is taking time.

7. Best Practices

7.1 Use a Non-Root Administrative Account

Perform normal administrative work using a non-root account and use **sudo** only when elevated privileges are required.

7.2 Use Strong Passwords

Use unique, difficult-to-guess passwords. Keep them private and never include them in screenshots or public GitHub repositories.

7.3 Keep the Server Updated

Regularly refresh package information and apply appropriate available upgrades.

7.4 Verify Services

Check important services after configuration. Do not assume a service is operational simply because its package was installed.

7.5 Record Network Information

Document the hostname, network interface, IP address, network method, and installed services, but never publicize passwords.

7.6 Install Only Required Software

Avoid unnecessary packages and services because they increase complexity and maintenance requirements.

7.7 Protect SSH Access

Enable SSH only when remote administration is needed and follow organizational security policies for authentication and access controls.

7.8 Back Up Important Data

Maintain appropriate backups before major system changes. VM snapshots can help during testing but should not automatically replace a proper backup strategy.

7.9 Document System Changes

Record what was changed, why it was changed, when it was changed, and how the system was configured.

7.10 Verify Before Deployment

Before declaring a server ready, verify login, hostname, networking, connectivity, updates, required services, and SSH status.

8. Quick Command Reference

Display Hostname

```
hostname
```

Purpose: Displays the current server hostname.

Display Network Information

```
ip addr
```

Purpose: Displays network interfaces and IP addresses.

Test Internet Connectivity

```
ping -c 4 google.com
```

Purpose: Sends four ping requests to test connectivity.

Update Package Information

```
sudo apt update
```

Purpose: Retrieves current package information.

Upgrade Installed Packages

```
sudo apt upgrade -y
```

Purpose: Installs available upgrades and confirms automatically.

Check SSH Status

```
systemctl status ssh
```

Purpose: Displays the current status of the SSH service.

Start SSH

```
sudo systemctl start ssh
```

Purpose: Starts the SSH service if installed but stopped.

Exit a systemctl Status Screen

```
q
```

Purpose: Returns to the command prompt.

9. References

- Canonical. Ubuntu Server Documentation. <https://documentation.ubuntu.com/server/>
- Oracle. Oracle VirtualBox User Manual. <https://docs.oracle.com/en/virtualization/virtualbox/>
- Microsoft. Windows Server Documentation. Microsoft Learn. <https://learn.microsoft.com/windows-server/>
- Rocky Linux Project. Rocky Linux Documentation. <https://docs.rockylinux.org/>
- UEFI Forum. UEFI Specifications and Resources. <https://uefi.org/specifications>

Conclusion

This manual provides a complete basic procedure for deploying Ubuntu Server using Oracle VirtualBox. It covers the installation process from virtual machine creation through network, storage, account, hostname, and OpenSSH configuration.

The verification procedures ensure that the newly installed server can successfully boot, authenticate users, obtain network information, access the internet, receive system updates, and run the SSH service.

Following a documented and repeatable installation process is important in system administration because it improves consistency, simplifies troubleshooting, and allows another administrator to understand and reproduce the deployment.